



SEEING THE UNSEEN

Data Governance, Trust, and Fraud Detection

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THE CHALLENGE

Fraud Drains Billions and Erodes Public Trust

\$233B–\$521B

Estimated annual federal fraud losses (3–7% of federal spending)

\$162B

In improper payments (FY 2024) — \$121B concentrated in just 5 programs, including Medicaid (\$31.1B)



AI-Accelerated Fraud

Criminals exploit automation and AI to scale fraud operations — generating fake claims, synthetic identities, and forged documents faster than manual review can keep up.



Siloed Data

Claims, eligibility, provider, and payment data live in disconnected systems — fraud hides in the gaps.



Speed Mismatch

Investigations take weeks. Fraudsters adapt in hours. Pay-and-chase recovery averages just \$0.20 on every dollar lost.

Source: GAO-26-108850, Jan. 2026



THE REALITY

Analysts Are Fighting Modern Fraud with Legacy Tools



Doesn't Scale

Legacy fraud systems rely on manual reviews and static rule sets. As claim volumes grow, investigative capacity stays flat — backlogs expand and fraudsters exploit the delay.



Only Finds the Familiar

Rules-based detection catches known patterns but misses novel schemes. If you haven't written a rule for it, you won't detect it — leaving billions in undiscovered fraud.



Blind to the Unknown

Synthetic identities, coordinated fraud rings, and AI-generated deepfakes are invisible to traditional tools. Without behavioral analytics and graph analysis, these threats go undetected.



THE OPPORTUNITY

What If You Already Had the Platform?

Elasticsearch powers security, observability, and search for thousands of public sector agencies. The same platform — the same data — can detect fraud.



Security / SIEM

What you know

Detection rules, threat intel, and automated response across your infrastructure.



Observability

What you see

Logs, metrics, and APM for full-stack visibility into application and system health.



Search / GenAI

What you find

Vector search, RAG, and generative AI to unlock knowledge across unstructured data.



Fraud Detection

What's next

Rules, ML anomaly detection, and graph analytics — powered by the same platform you already run.

One platform. One data store. One skill set. Extend what you already have into fraud detection — no new stack required.



THE FRAMEWORK

A Layered Approach to Fraud Detection

Three detection tiers — from deterministic rules to machine learning to graph analytics.

- 1 Known-Knowns
- 2 Known-Unknowns
- 3 Unknown-Unknowns
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Known-Knowns

Detection Rules + ES|QL — catch the fraud you already understand

CLAIMS PIPELINE

CLM-4401	Dr. Smith	\$2,340	✓
CLM-4402	ABC Clinic	\$18,750	⚠
CLM-4403	Dr. Jones	\$890	✓
CLM-4404	XYZ Labs	\$45,200	⚠
CLM-4405	Dr. Patel	\$1,100	✓
CLM-4406	Quick Care	\$32,800	⚠

ES|QL DETECTION QUERY

Rules matched — building query...



LIVE DEMO

Fraud Detection with Elastic

Medicaid Provider Spending — open-source federal data



ES|QL Queries

Explore Medicaid provider spending with piped, composable queries directly in Kibana.



Machine Learning

Anomaly detection jobs surface outliers in billing patterns, geography, and timing.



Graph Exploration

Map provider-beneficiary relationships to expose coordinated fraud rings.



Workflow Automation

Trigger investigations, flag claims, and route cases — all from detection rules.

Let's See It in Action

